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SCHOLARLY REVIEW

Contaminants in Dietary Supplements: Toxicity, Doping Risk, and Current Regulation

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Athletes, both amateur and professional, often resort to the consumption of nutritional supplements without professional supervision and without being aware of the risks they may entail. We conducted an exhaustive literature search to determine the most common substances found as contaminants in dietary supplements. For each substance, we analyzed its mechanism of action, clinical indication, health risk, and putative use as doping agent. In addition, we evaluated the current regulation of these supplements. Contamination of nutritional supplements (accidental or intentional), especially with steroids and stimulants, is a hazardous situation. The prolonged consumption of these products without being aware of their composition can cause serious health risks and, in the case of professional athletes, a possible sanction for doping.

***Keywords**:* diet, ergogenic aids, steroids, stimulants, health risk, sports

Since ancient times, performance in sports has been tried to be improved by employing a variety of strategies, including the use of ergogenic aids and dietary supplements (Jovanov et al., 2019). In athletes, the use of supplements is far more common than in the general population, with reported figures of users of 44%–100% depending on sport, age, and sex (Maughan et al., 2007). However, the ingestion of an inappropriate product at the wrong dosage or of those with a dubious origin can lead to serious health problems (Porrini & Del Bo', 2016). In addition, dietary supplements are a major cause of antidoping rule violations. A recent study shows that in 26% of positive antidoping tests between 2003 and 2020, the consumption of supplements was claimed as the source of the banned substance, although evidence to this claim was only found in half of the cases (Lauritzen, 2022).

The problem of the presence of contaminants in dietary supplements is currently on the rise. Studies have shown that about 15%–20% of perfectly legal nutritional supplements are contaminated, and that figure only refers to contamination with anabolic steroids (Geyer et al., 2008; Parr et al., 2011).

Since the implementation of legal frameworks for supplements both in the European Union (EU) and in the United States, with the Dietary Supplements Health and Education Act, the sales of these products have significantly increased (Starr, 2015). However, it has been from the 2010s onward, with new consumer profiles, when the expansion of the industry has skyrocketed. This new type of consumer profile is associated not only with a greater concern for health in the general population but also with the wider access to the internet provided by mobile devices (Płaciszewski et al., 2022). Dietary supplements may now be purchased rapidly and conveniently, but, at the same time, it is more likely that the customer is unaware of the deficient guarantee in the product formulation, contents, or effects. This becomes a matter of concern when 150 million people in the United States alone reportedly use dietary supplements, most of them daily (Starr, 2015), and the estimated growth of the global market for these products will reach USD 128.64 billion by 2028 when it was USD 71.81 billion in 2021 (Europe Dietary Supplements Market Size, Trends | Analysis [2020–2027], n.d.). A large study has shown that the most frequent contaminated products are marketed for sexual enhancement (45.5%), weight loss (40.9%), and muscle building (11.9%), with 20% of adulterated supplements containing more than one unapproved ingredient (Tucker et al., 2018).

In this narrative review, we intend to summarize the most frequent contaminants found in dietary supplements (due to both accidental and intentional contamination), what their putative effects can be, and, especially, to raise awareness on the dangers of taking these products without proper guidance.

Definition and Classification

Although we will refer to "dietary supplements" throughout the manuscript, the legally correct term in the EU is "food supplements." This term is defined by EU directive 2002/46/EC as foodstuffs intended to supplement the normal diet and consisting of concentrated sources of nutrients or other substances having a nutritional or physiological effect, in single or combined form, marketed in dose form, that is, capsules, tablets, pills and other similar forms, sachets of powder, ampoules of liquid, dropper bottles, and other similar forms of liquids and powders to be taken in small unit quantities.

On the other hand, "ergogenic aid" is a broader term. This concept is defined as any technique or method (nutritional, physical, mechanical, psychological, or pharmacological) carried to increase the capacity to perform physical work and improve performance. Thus, a dietary supplement can be considered an ergogenic aid (Porrini & Del Bo', 2016). However, it is difficult to impose a single or unitary definition of "dietary supplement" or "food supplement," even if it is as precise as it is in the United States or the EU. The reason is the lack of global consensus on which products fall within this definition, which leads to large discrepancies between countries in defining and regulating these supplements (Dwyer et al., 2018).

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Dietary supplements have been classified and categorized in many ways, but none is entirely satisfactory. Again, the lack of a global consensus and the great diversity of the groups make classifying the different supplements a challenging mission (Garthe & Maughan, 2018). The Australian Institute of Sport has created a general classification of sports food and supplements (https://www.ais.gov.au/nutrition/supplements), which we summarized in Table 1 and that may be illustrative. Of note, substances may move from one category to another depending on the evidence available at the time.

A rapid information system on foodstuffs (including dietary supplements) is in place in the EU. The Rapid Alert System Feed and Food (RASFF) provides a very useful tool in the detection of hazardous dietary supplements. The RASFF software application is the most efficient way to obtain information on different food products (https://webgate.ec.europa.eu/rasff-window/screen/search).

The World Anti-Doping Agency (WADA) annually establishes a list of prohibited substances and methods (https://www.wada-ama.org/en/prohibited-list) that threaten both the health of athletes and the ethical values of sport. As shown later, any of the products on the list can be found in a dietary supplement.

Current Regulation

Although there is a clear legislative framework within the EU and other regions for the use of dietary supplements, the aforementioned lack of global consensus creates many problems. A substance that in one country is freely included as a component in supplements may be regulated and subject to health supervision in other countries. For instance, dehydroepiandrosterone is not considered an anabolic steroid as defined by the U.S. Drug Enforcement Administration; however, WADA includes it on the prohibited list under the section of anabolic–androgenic steroids (AAS; Drug Enforcement Administration [DEA], Department of Justice, 2009). This situation is not helped by the position of countries such as China or India, with particular regulations heavily influenced by cultural roots, where traditional medicine or phytomedicines have a great importance.

There exists a legislation on supplements as foodstuffs within the EU as well as on those vitamins and minerals that can be part of them. There are also a number of directives established by the European Commission to try to protect consumers and avoid health risks, and to this end, it has drawn up a list of substances that can potentially cause adverse health effects. The main European standards for nutritional supplements are listed here.

- Directive 2002/46/EC: Initial regulation on the use of food supplements, establishment of a list of vitamins and/or minerals used in supplements, and additional labeling requirements. Consideration of supplements as foodstuffs.
- Regulation (EC) No. 1925/2006 on the addition of vitamins and minerals and of certain other substances to foods.

These two first regulations have been later amended by Regulation (EC) No. 1170/2009.

- Regulation (EC) No. 178/2002: Attribution of responsibility for the safety of the supplement to the food business operator placing the product on the market.

Table 1 General Classification of Sports Food and Supplements by the Australian Institute of Sport

Evidence level and use within supplement programs	Category	Application	Examples
A. Strong scientific evidence for use in specific situations using evidence-based protocols	Sports foods	Specialized and useful products for when it is impractical to guarantee a source of nutrients with common and usual foods	Gels, sports drinks, bars, sports confectionary, protein powders, etc.
	Supplements for medical use	Used to address issues of clinical use, such as diagnosed nutritional deficiencies	Multivitamin supplements, minerals, iron, calcium, vitamin D supplements, probiotics, etc.
	Performance supplements	Intended to legally increase the performance of athletes	Caffeine, β-alanine, protein, creatine, bicarbonate, nitrate, and combinations thereof
B. Emerging scientific support, considered for use within research or clinical monitoring situations	Antioxidants, polyphenols, tastants, and others	Intended to legally increase the performance of athletes	Collagen, curcumin, carnitine, ketone, vitamin C, fish oils, menthol, fruit-derived polyphenols, *N*-acetyl cysteine
C. Scientific evidence not supportive of benefit for athletes or no research undertaken. Not advocated for general use	Named products	Used for a variety of functions and health-related issues	Branched-chain amino acids, magnesium, phosphate, vitamin E, prebiotics, tyrosine, and alpha lipoic acid
	Products in A and B groups used outside approved protocols		
	Ingredients not found in A, B, or D groups likely fall within this category		
D. Banned or at high risk of contamination with substances that could lead to a positive doping test. Not to be used by athletes	Banned substances (see WADA list)	Intended to illegally increase performance in athletes	Stimulants, prohormones, AAS, GH and peptide releasers, b2 agonists, metabolic modulators, SARMs, and other (colostrum)

Note. WADA = World Anti-Doping Agency; SARM = selective androgen receptor modulator; GH = growth hormone; AAS = anabolic androgenic steroids.

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- Regulation (EC) No. 1924/2006: Evaluation of substances added to foods regarding nutrition and health claims.
- Regulation (EC) No. 1333/2008: It establishes a list of additives authorized for use in food supplements: sweeteners, colorings, and so on.
- Regulation (EC) No. 2015/2283: Evaluation of substances with no significant history of consumption as food in the EU before 1997.
- Regulation (EU) No. 2023/915 (previously 1881/2006): It regulates the limits for contaminants and other components in foodstuffs.

The responsibility for the safety of food supplements lies with the manufacturer, importer, supplier, or distributor as laid down in Regulation (EC) No. 178/2002. However, EU member states may request the notification of the marketing of each supplement and the authority to control its use once the product is in circulation on their territory. In any case, it is important to stress that, despite the aforementioned set of regulations, there is a shocking lack of enforcement of this legislation by the different agencies, which creates opportunities for rogue companies to market dubious products. In practice, there is no guarantee that these supplements are free of potentially hazardous contaminants, and only products that have been shown to be unequivocally harmful to the consumer are withdrawn. Numerous enforcement challenges remain, including an adequate evaluation of safety and efficacy, proper labeling oversight, limiting marketing unfounded claims, or controlling the report of adverse events (Starr, 2015).

Contaminants in Supplements: Most Frequent Substances

Although any of the various substances listed by WADA as doping substances could appear in a dietary supplement, there are some that are more commonly found in these preparations. Most of the supplements withdrawn from the market due to contamination, which has been shown to be frequently intentional, include: AAS, central nervous system stimulants, and/or diuretics (Mathews, 2018). All of these are substances listed by WADA as banned both for their performance-enhancing capabilities and for the health concerns associated with their nonprescribed use. With regard to these health issues, it should be remarked that even though the presence of these contaminants in dietary supplements may constitute a serious risk, the adverse effects described in this section would presumably be more frequent and serious in outright doping (Geller et al., 2015).

Anabolic–Androgenic Steroids

AAS are synthetic derivatives of male sex hormones (mainly testosterone). Those that have not been tested for use in humans are known as designer AAS (Mathews, 2018). These compounds increase protein synthesis by interacting with intracellular nuclear androgen receptors, triggering the development of male secondary sexual characteristics through gene expression (Bond et al., 2022). They are used in the treatment of various diseases, such as anemia, terminal and debilitating diseases, metastatic breast cancer, replacement in testicular failure, or hypogonadism (Bond et al., 2022). They are also used in nonmedical conditions by athletes for performance-enhancing purposes because of the obtained increase in muscle size produced by the boost of protein synthesis mainly due to the increase in anabolic properties. Adverse effects, which are obviously more likely to occur when AAS have been abused, include decreased secretion of gonadotropins and, therefore, of endogenous testosterone, cholestatic jaundice, psychotic symptoms, aggressiveness, irritability, kidney and liver disease, coronary heart disease, and cardiovascular events (Albano et al., 2021; Nieschlag & Vorona, 2015). In addition, there also are sex-specific adverse effects: gynecomastia and sterility in men or deeper voice, hirsutism, and inhibition of ovulation in women. These sex-related effects are very much decreased when designer steroids are used, which are manufactured to maximize their anabolic properties while keeping androgenic effects at bay. Among the most frequently found AAS in dietary supplements are stanozolol, dehydroepiandrosterone, dihydrotestosterone, oxandrolone, androstenedione, methandienone, methasterone, or metabolic precursors, such as prostanozol (Martínez-Sanz et al., 2017; Mathews, 2018; Walpurgis, Thomas, et al., 2020).

The detection of designer AAS in dietary supplements is becoming a matter of concern. For instance, dymethazine and methylstenbolone have been identified in products marketed for vitamin B supplementation and general nutritional purposes, respectively (Cavalcanti et al., 2013; Rahnema et al., 2015). Their consumption has been related to liver damage like severe hepatoxicity, cholestasis, carcinoma, and other adverse effects, such as acute renal failure. Both of these substances are alkylated, which increases their bioavailability and toxicity and were not identified by name in the product labels (Cavalcanti et al., 2013; Rahnema et al., 2015). In the same line, a large study on bodybuilding dietary supplements yielded significant results. Twenty-four supplements from fitness shops in the United Kingdom were tested for possible banned substances. Of them, 23 were contaminated by different steroids, many of them not tested on humans, namely dehydroepiandrosterone, methyl-1-testosterone, androstenedione, desoxymethyltestosterone, methasterone, androst-4-eno-3,11,17-trione, furazabol, reduced desoxymethyltestosterone, 19-norandrosta-4,9-dien-3,17-dione, 4-chloro-17a-methyl-androst-4-eno-3,17-b-diol, adrenosterone, methasterone, and 6-bromo-androstenedione. In addition, 16 of the 23 supplements were found to be incorrectly labeled (Abbate et al., 2015).

Central Nervous System Stimulants

These are stimulant drugs that can powerfully excite the central nervous system through increase of catecholamines release (noradrenaline and dopamine) and/or inhibition of their reuptake. This results in higher levels of these neurotransmitters, thus leading to increased alertness, locomotor activity, and euphoria (Docherty, 2008; Heal et al., 2014). Some of them are used to treat attention-deficit hyperactivity disorder in children, for example, methylphenidate (Sharma & Couture, 2014). Their use in doping is based on the reduction of physical and/or mental fatigue and a small, but significant, increase in sporting performance, especially in endurance sports (Docherty, 2008; Stull et al., 2021). Adverse effects include tachycardia, anxiety, panic, hallucinations, increased aggression, amphetamine-related psychosis, and cardiovascular events (Docherty, 2008; Heal et al., 2014). Stimulants are often found in supplements marketed as "fat-burners" due to their thermogenic effect that increases fat metabolism. Some examples are sibutramine, ephedrine, dimethylamylamine (DMAA) and other amphetamine derivatives, phenylethylamine and derivatives, and lorcaserin (Martínez-Sanz et al., 2017; Mathews, 2018; Walpurgis, Thomas, et al., 2020).

One of the most commonly found stimulants in dietary supplements is DMAA, an amphetamine derivative. For example, a German study detected its presence in 16 supplements marketed

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as weight loss products seized during customs controls, with no or incomplete mention in the product label (Monakhova et al., 2014). DMAA has been associated with amphetamine-like symptoms and even deaths.

Recently, traces of a comparatively less common stimulant, mitragynine, have been found in herbal supplements and several energy drinks (Arndt et al., 2011; Guddat et al., 2016). Mitragynine is obtained from kratom plant. In Asia, kratom leaf has traditionally been used to increase physical workload and to induce euphoric effects. There are reported cases of death from kratom misuse and its combination with other drugs (21). In the same manner, ephedrine alkaloid dietary supplements have been associated with more reports of death, myocardial infarction, and other serious events than all other dietary supplements combined (Charatan, 2003). As a result, the U.S. Food and Drug Administration banned the sale of products containing ephedrine alkaloids in 2004.

As in the case of AAS, designer stimulants are being detected in dietary supplements. A dietary supplement called NOXPUMP, promising a boost in strength, energy, endurance, and motivation, was shown to be contaminated with two new amphetamine analogues: *N,N*-dimethyl-2-phenylpropane-1-amine and 1-amino-2-phenylpropane (Kwiatkowska et al., 2015). A large study by Cohen et al. (2021) aimed to determine the presence of new designer stimulants in dietary supplements. Thirty-five weight loss supplements from 17 different brands purchased online were analyzed. The following designer stimulants were found: deterenol, 1-amino-2-phenylpropane, phenpromethamine, octodrine, oxilofrine, higenamine, 1,3-DMAA, 1,4-DMAA, and 1,3-dimethylbutylamine. Many of these drugs have never been tested in humans and have been associated with cardiac arrest, stroke, and sudden death.

Diuretics

Thiazide diuretics and loop diuretics, for example, hydrochlorothiazide or furosemide, are those most commonly found in dietary supplements (Favretto et al., 2019; Mathews, 2018). Their mechanism of action is based on the increase of the excretion of different electrolytes in the kidney and/or the inhibition of their reuptake. To decrease the concentration of electrolytes in urine, the kidney increases the amount of water, thereby increasing diuresis (Wile, 2012). These drugs are vastly used clinically in hypertension, nephrotic syndrome, acute pulmonary edema, or renal failure (Blowey, 2016; Wile, 2012). Their use as doping agents is explained by their ability to increase the volume of water in urine, which results in the decrease of the concentration of other banned substances below the detection threshold. It is also illegally used in sports when a rapid weight loss is needed, for example, in sports with weight divisions, such as boxing, wrestling, or weightlifting, and in sports and activities that require low body weight, such as gymnastics or ballet (Martínez-Sanz et al., 2017; Walpurgis, Thomas, et al., 2020). The possible adverse effects are multiple; particularly relevant are arrhythmias and heart failure due to the loss of potassium, which is essential for contractile tissues (potassium-sparing diuretics are often used), dehydration, electrolyte disturbances, lower blood pressure, and dizziness (Stull et al., 2021; Wile, 2012). The presence of diuretics has been confirmed in dietary supplements consumed by professional athletes who tested positive in doping controls. These drugs were not listed in the product label (Favretto et al., 2019).

In addition to these three pharmacological groups, there are others that are less frequently found in dietary supplements but that have also been associated with serious adverse effects.

Benzodiazepines

Benzodiazepines are compounds with hypnotic, anxiolytic, and central muscle relaxant properties that enhance the effects of gamma-aminobutyric acid, a neurotransmitter that reduces neuronal excitability. Adverse effects include acute toxicity, dizziness, drowsiness, dependence, tolerance, ataxia, and confusion. Its application in doping is limited, being a substance not banned by WADA but with relevant side effects on health (Nielsen, 2017). A rapid methodology has been developed to identify these substances in dietary supplements that were tested in 45 batches of supplements from 20 different brands. Three of the products were contaminated with clonazepam, diazepam, and alprazolam without label reference. Furthermore, adverse effects such as drowsiness, dizziness, weakness, and respiratory depression were reported after consumption of the products (Jiang et al., 2015).

Glucocorticoids

Glucocorticoids are steroid hormones widely used for the treatment of various diseases, such as inflammatory and autoimmune disorders, Addison's disease, adrenal hyperplasia, asthma, and so on. Their use in doping is controversial; on the one hand, they increase adrenal androgens (with far weaker effect than AAS), but on the other hand, they stimulate proteolysis and decrease glucose uptake by muscle. Their adverse effects are more likely to result from prolonged long-term use, the most important of which is suppression of the hypothalamic–pituitary axis that results in the inability to generate endogenous corticosteroids that are necessary for homeostasis (Kapugi & Cunningham, 2019). Some supplements have been found to be contaminated with hydrocortisone, a substance banned by WADA (Li et al., 2018).

Selective Estrogen Receptor Modulators

Selective estrogen receptor modulators are drugs used in estrogen receptor-positive breast cancer. Their mechanism involves blocking estrogen receptors in the hypothalamus–pituitary axis, thus overriding the negative feedback on testosterone production. This indirect increase in testosterone provides the athlete with the anabolic effects of AAS without the androgenic effects. However, their use has been linked to thromboembolic problems, deep vein thrombosis, musculoskeletal disorders, nausea, hot flushes, arthritis, and weakness (Haskell, 2003; Park et al., 2021). One of the main selective estrogen receptor modulators is tamoxifen, which has been identified in supplements advertised as muscle enhancers (Evans-Brown et al., 2014).

New Anabolic Agents

Recently, some dietary supplements have been found to be contaminated with new drugs under development: selective androgen receptor modulators (SARMs). These types of ligands act as a full agonist at anabolic organ receptors (muscle and bone) but as a partial agonist at androgenic tissue receptors, resulting in an increase in the individual's anabolic characteristics without the presence of androgenic effects. This makes SARMs useful for treating diseases such as hypogonadism, osteoporosis, AIDS, and various anemias but, at the same time, makes them very attractive as doping agents as the androgenic actions of classic anabolic agents are canceled out (Chen et al., 2005). Two of the most utilized new anabolic agents are ligandrol and ostarine. Ostarine in particular, due to its apparent lack of serious side effects, has become the

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most relevant SARM in doping in recent years; however, there is a reported risk of cardiac events and hepatotoxicity. Traces of ostarine have been found in products used as dietary supplements (Walpurgis, Rubio, et al., 2020).

Growth Hormone-Releasing Peptides

Growth hormone (GH)-releasing peptides are proteins that stimulate the release of GH in the adenohypophysis. The most frequently found contaminant in supplements is the GH-releasing peptide 2. These peptides, banned by the WADA, are agonists of the ghrelin receptor, thus inducing the release of GH in the pituitary. Insulin-like growth factors produced in response to increased GH, especially insulin-like growth factor-1, are responsible for the anabolic effects demanded by doping. In clinical practice, GH is used in GH deficiency (pituitary dwarfism) or in disorders of insulin-like growth factor formation or action. In addition to acromegaly, its use can cause dizziness, hot flushes, diabetes, hypertension, joint pain, and edema (Argente et al., 1996). Thomas et al. (2010) developed a routine assay for the detection of these peptides in dietary supplements and found GH-releasing peptide 2 in one of the tested products. Interestingly, its presence was acknowledged in the label but not by name, just in the form of chemical formulation (with an incorrect amino acid sequence) and with no indication of the amount per tablet (Thomas et al., 2010).

Erythropoiesis-Stimulating Factors: Cobalt

Cobalt is a mineral that, in its ionic form (Co^{2+}), stabilizes the hypoxia-inducible factors 1- and 2-alpha that favor an increased production of erythropoietin. Erythropoietin stimulates the production of erythrocytes, thereby increasing the amount of circulating oxygen. This is the origin of its use in doping, conferring a clear advantage, especially in aerobic sports such as cycling and long-distance running. Between the 1940s and 1970s, treatment with high doses of cobalt was widely used to deal with different types of anemias, but this practice was abandoned due to the significant increase in hemoglobin, hematocrit, and reticulocytes and its relationship with acute intoxication, organ damage, gastrointestinal disease, thyroid dysfunction, myocardial effects, and neuronal and sensory alterations. In equestrian sports, cobalt doping is an issue of crucial importance that has extended to human sport, so much so that cobalt was included as a banned substance in 2015 by WADA. Cobalt has been found in dietary supplements that promised to stimulate erythropoiesis and improve performance in endurance sports (Schmidt et al., 2019). A study in Germany and Switzerland analyzed 19 supplements obtained online and by confiscation. Eleven of the products were contaminated with cobalt and nickel. Nine of these did not mention these ions in the label, and in the other two, their detected levels were higher than indicated (Thevis et al., 2016). Even though it has not yet been proven that supplements contaminated with cobalt may produce an enhanced athletic performance, the concentrations detected by Thevis et al. (2016) would be more than enough to cause a notable erythropoietin elevation and a subsequent performance improvement (Hoffmeister et al., 2018).

Types of Contamination and Contract Manufacturing

Contamination of dietary supplements is generally difficult to assess. It is established that it may be intentional, by adding an undeclared component that does not appear on the labeling, or accidental, by cross-contamination. Intentional contamination is the malicious and conscious introduction of foreign substances into food and, thus, into food supplements (Akahane et al., 2023). Cross-contamination is the process by which microorganisms or other substances foreign to the product being manufactured or marketed are unintentionally transferred from one food or object to another food, with harmful effects. A few years ago, the highest prevalence of contamination was due to cross-contamination. It was mainly caused by poor cleaning of containers used for transport and storage of these supplements, especially when other pharmacologically active products were manufactured on the same production line. This was the case, for instance, of prohormone preparations (Judkins et al., 2010; Judkins & Prock, 2012). Some manufacturers of prohormones, legally marketed in the United States, also manufactured dietary supplements, which led to cases of contamination (Geyer et al., 2008). The situation was complicated by the fact that some of the sources of cross-contamination were not the production line itself but the prior contamination of the components (Geyer et al., 2008; Judkins et al., 2010). It is, therefore, paramount that the company responsible for manufacturing dietary supplements follows what is known as current good manufacturing practice (cGMP), a term initially coined in the United States Center for Food Safety and Applied Nutrition, (2022). The use of cGMPs would largely prevent this cross-contamination. Companies must be committed to cGMP compliance: to educate themselves, conduct internal audits, use appropriate criteria and qualified consultants, and above all, implement appropriate corrective measures (Whitsitt et al., 2013). However, it should be stressed that the cGMP certification does not fully guarantee the absence of substances with the potential to cause an adverse doping test result. cGMPs certify that the manufacturing process has been carried out correctly, which, it is true, minimizes the possibility of contamination. However, no tests aimed to identify doping agents in the supplements are conducted. In addition, although the cGMP seal is well established in the United States, its use is far less common in other regions. These limitations of the cGMPs in terms of the use of supplements by professional athletes explain, in part, the surge of third-party companies created to verify the presence of doping agents in dietary supplements (see the following section for more information on this).

There presently seems to be a growing trend toward the deliberate contamination of dietary supplements with synthetic pharmaceuticals and analogues to enhance their effectiveness and increase sales subsequently (Amidžić et al., 2023). These non-registered analogues of approved drugs, which are sometimes added to make detection more challenging, represent the most serious concern as they have not been clinically tested, and hence, unexpected adverse effects may occur (Muschietti et al., 2020).

Of Interest to Athletes and Coaches

In addition to the risks associated with the contamination of dietary supplements, a different but important question is whether the consumption of these products is really necessary in certain cases. The International Olympic Committee established a decision tree to guide both amateur and professional athletes (who may or may not have a nutritional deficiency) when taking supplements or ergogenic aids. Similarly, the International Olympic Committee's own consensus also points out the need to seek the advice of an appropriate doctor or nutritionist when these products are really necessary. Based on these International Olympic Committee's guidelines, Figure 1 shows a diagram that deals with different

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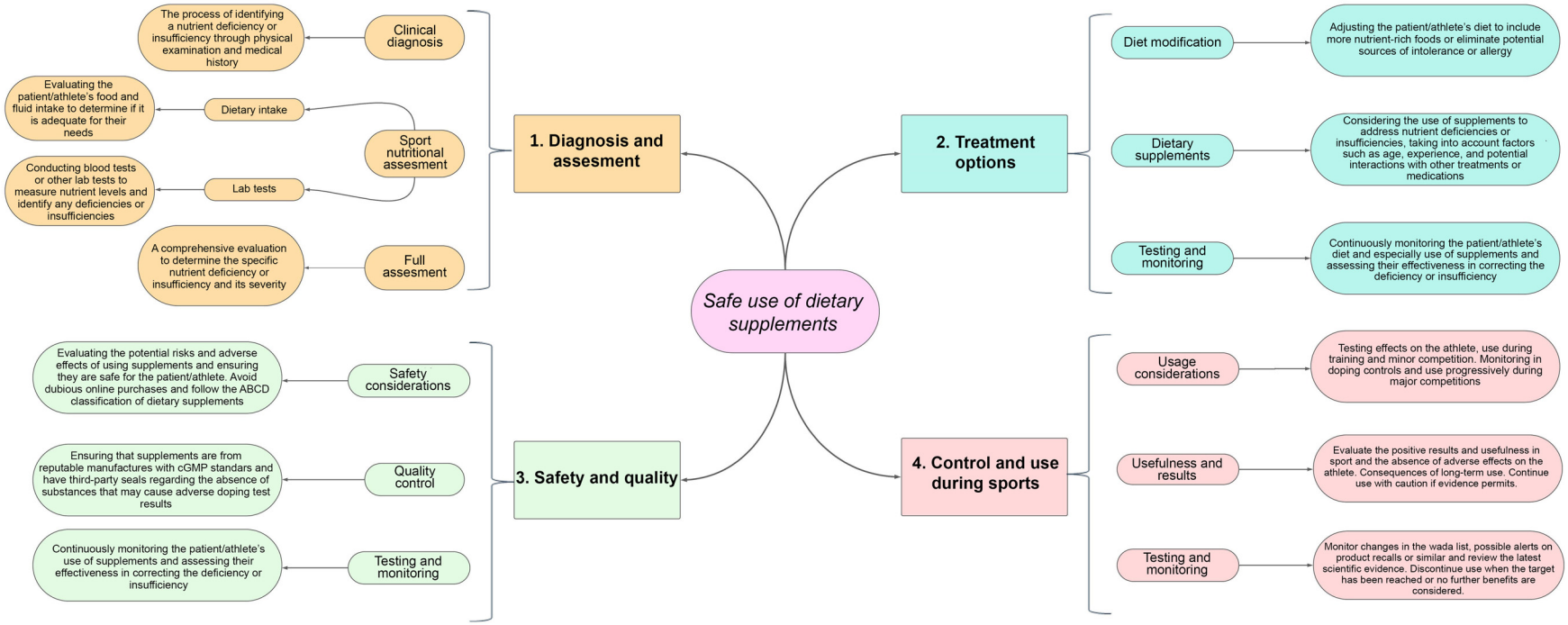

Figure 1 — Diagram showing four different considerations (diagnosis and assessment, safety and quality, treatment options, and control and use during sports) to take into account when consuming dietary supplements. cGMP = current good manufacturing practice.

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Figure 2 — Certification companies issuing seals of approvals for dietary supplements.

aspects on how and when to use dietary supplements, namely diagnosis and assessment, safety and quality, treatment options, and control and use during sports.

As WADA indicates, failure to follow these institutional and professional recommendations may result in involuntary doping and/or serious health consequences. Importantly, WADA states that for an adverse doping test result, the responsibility rests with the athlete. To avoid this mishap, WADA publishes on its website information on the different supplements that have tested positive in its accredited laboratories. However, WADA does not endorse or participate in any supplement certification process, let alone in their manufacture, even though some companies claim that their supplements or products have been certified and/or approved by WADA. This agency may test commercial material only when requested by an antidoping department or organization as part of an investigation into a positive doping case. In any case, although WADA does not officially endorse them, it does recognize the existence of third-party companies that evaluate and test the safety of various of these dietary supplements. These testers greatly reduce the risk of contamination. Some of these companies are the *Netherlands Anti-Doping Authority Project* in the Netherlands, the *Cologne List* in Germany, *Informed Sports* in the United Kingdom, *Consumer Labs* in the United States and Canada, *NSF International*, and *Aegis Shield*. Many of these companies have "seals of approval" (Figure 2) to be placed on the label of those supplements that have been evaluated for quality, purity, and composition.

It is of the utmost importance that both athletes and coaches are aware that the prolonged consumption of these products without strict medical advice can cause serious health risks and a possible sanction for doping. Despite the aforementioned legislation, the online supplement/medical business is presently out of control, and athletes competing under the code should probably never use online ordering at all. To make matters worse, many studies have shown that most athletes are unaware that dietary supplements can have adverse effects and do not know the active ingredients or the recommended doses (Mathews, 2018).

Conclusions

The use of dietary supplements has exponentially increased in the last years due to a more specialized approach to sports and, in the case of nonprofessional athletes, to an increased awareness about nutrition and well-being. Available evidence shows that a non-negligible percentage of these supplements are contaminated with potentially harmful substances, the most frequent of which are AAS and stimulants. Of particular concern is the development of designer agents included in these groups whose pharmacological and adverse effects have, as yet, not been tested in humans. In addition, diuretics also play an important role in the contamination of dietary supplements. Indeed, national antidoping agencies continuously inform us about numerous products and supplements being removed from the market because they incorporate undeclared diuretics among their active ingredients. Another category of contaminant that is experiencing an exponential rise is SARMs. SARMs are becoming common substances in different products marketed for the bodybuilding world due to their low adverse effects because of the blockade of androgenic action. Again, numerous products have been removed from the market for containing this new type of banned substance. Less frequent are contaminants such as peptide hormones, aromatase inhibitors,

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glucocorticoids, or benzodiazepines. The latter deserve special mention as their consumption has greatly increased in recent years due to the global pandemic context. It is interesting to know that these substances can also be inadvertently consumed through dietary supplements.

In summary, contamination of dietary supplements may occur, and the probability of taking a contaminated product is higher depending on the area of sale (particularly in online shopping) and the type of product sold (athlete oriented, weight loss, etc.). This contamination is a major problem that needs to be solved on an international basis. There is a need for common cooperation between different countries and bodies to enforce legislation and to curb the ability of dishonest companies to continue to place contaminated products on the market. The present lack of common criteria and actions contributes to the maintenance of doping practices in professional athletes as well as exposes a more sedentary population, and/or young people who see these supplements as a way to maintain an ideal body image, to unnecessary risks.

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